

P2P Pricing Mechanism: Comparative Results

Across Community Sizes, Compositions, and Stakeholder Priorities

Simulation: 30 days, 15-min resolution, $\lambda_{\text{buy}} = \text{Rs.}6.0/\text{kWh}$, $\lambda_{\text{sell}} = \text{Rs.}3.4/\text{kWh}$, $\alpha = 0.12$

Table 1: Savings Breakdown — Consumer vs. Prosumer

Each scenario shows Total / Consumer / Prosumer savings (%) and Fairness Index for all three mechanisms.

Community	N	MMR				Bill-Sharing				SDR + DSM				Best
		Tot	Con	Pro	$\mathcal{F}$	Tot	Con	Pro	$\mathcal{F}$	Tot	Con	Pro	$\mathcal{F}$	
SDR wins: Consumer-heavy — aggressive DSM offsets low PV with demand flexibility														
7C+1SP+1MP+1LP	10	8.6	5.6	26.4	.35	7.5	13.8	−31	.00	10.1	5.6	37.2	.26	SDR
5C+2SP+2MP+1LP	10	7.9	6.5	11.0	.74	6.2	10.8	−3.8	.00	11.5	5.7	24.1	.38	SDR
25C+5SP+5MP+5LP	40	8.0	5.7	17.3	.50	6.8	12.5	−16	.00	10.0	5.4	28.6	.32	SDR
30C+10SP+5MP+5LP	50	7.2	5.4	12.8	.59	6.0	11.4	−11	.00	9.1	5.0	22.1	.37	SDR
MMR wins: Balanced/diverse — best fairness-savings trade-off														
1C+2SP+2MP	5	5.8	7.0	5.2	.85	3.2	5.3	2.3	.61	13.1	5.9	16.2	.53	MMR
10C+8SP+7MP+5LP	30	8.4	7.6	9.2	.90	5.9	9.2	2.4	.41	14.9	7.0	23.5	.46	MMR
10C+15SP+15MP+10LP	50	8.1	8.1	8.1	1.0	4.9	6.7	3.9	.73	18.3	7.7	24.2	.48	MMR
Bill-Sharing wins: Prosumer-dominated — only mechanism with equitable distribution														
1C+1SP+1MP+2LP	5	12.1	9.0	15.2	.74	7.8	8.1	7.6	.97	26.3	8.6	43.6	.33	BS
1C+2SP+3MP+4LP	10	10.6	9.4	11.0	.92	5.1	4.9	5.1	.97	29.8	9.1	37.2	.39	BS
5C+5SP+10MP+10LP	30	10.2	9.0	11.0	.90	5.8	6.3	5.5	.93	25.5	8.7	35.6	.39	BS
3C+2SP+5MP+10LP	20	15.4	9.9	20.7	.65	8.2	6.0	10.3	.74	41.3	9.8	71.8	.24	BS

Tot = community-wide savings (%) vs. conventional. **Con/Pro** = group-specific savings. $\mathcal{F}$ = fairness index. Negative Pro = prosumers pay *more* than conventional. Winner by weighted score (cost 40%, fairness 30%, stability 30%). SDR achieves highest raw savings in *every* scenario but often loses on fairness — prosumers capture 3–8× more benefit than consumers.

Table 2: How Stakeholder Priorities Change the Optimal Mechanism

Table 3: Decision Guide

Community	N	Cost-first (70/15/15)	Balanced (40/30/30)	Fairness-first (20/60/20)
7C + 1SP + 1MP + 1LP	10	SDR	SDR	MMR
30C + 10SP + 5MP + 5LP	50	SDR	SDR	MMR
10C + 15SP + 15MP + 10LP	50	SDR	MMR	MMR
10C + 8SP + 7MP + 5LP	30	SDR	MMR	MMR
1C + 2SP + 2MP	5	SDR	MMR	MMR
5C + 5SP + 10MP + 10LP	30	SDR	Bill-Sharing	Bill-Sharing
1C + 2SP + 3MP + 4LP	10	SDR	Bill-Sharing	Bill-Sharing
3C + 2SP + 5MP + 10LP	20	SDR	Bill-Sharing	Bill-Sharing

Weight triplets: (cost / fairness / stability). SDR *always* wins when cost is the dominant priority — its DSM-enhanced savings (10–41%) far exceed other mechanisms. But when fairness enters the scoring, MMR or Bill-Sharing take over because SDR’s benefits are inequitably distributed.

Mechanism	Best Suited For	Community Profile	Key Trade-off
SDR + DSM	Any community where maximum cost reduction is the primary goal	Consumer-heavy (limited PV but flexible loads)	Highest savings by far; prosumers gain 3–8 $\times$ more than consumers ($\mathcal{F} = 0.24\text{--}0.53$)
MMR	Balanced or diverse communities seeking fair cost sharing	20–50% consumers; mixed PV sizes	Nearly equal group savings (Con $\approx$ Pro); $\mathcal{F} = 0.85\text{--}1.0$; robust across compositions
Bill-Sharing	Prosumer-dominated communities needing guaranteed equity	$C \leq 15\%$; many large PV systems	Consumer and prosumer savings nearly identical ($\mathcal{F} > 0.93$); never harms prosumers

No single mechanism dominates. SDR maximizes total savings but creates inequity. MMR balances savings and fairness. Bill-Sharing guarantees equity at the cost of lower total savings. The optimal choice is a *policy* decision driven by community values.